

Gender Bias in Yearbook Comments

Data and Samples

Raw data
38,993 → **After quality filters**
37,553 → **After deduplication**
37,476

Definition	Included comment types	<i>N</i>	Female	Female share
Text-focused sample	Biography, short comment, mixed, other, quotation/maxim, caption/fragment	36,826	8,887	24.16%
Full sample	Text-focused sample plus basic-profile records	37,476	8,979	23.98%

Ratios use original school files. Filters removed 1,440 records; deduplication removed 77 records.

[Online data viewer](#)

Comment Types

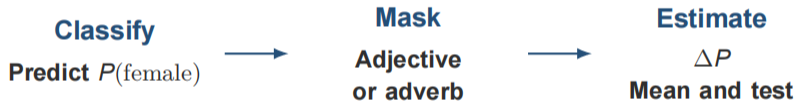
Type	<i>N</i>	Example
Biographical sketch	15,958	Great student of the Bible. Prefers the revised version.
Short comment	5,921	One of our blooming businessmen.
Mixed profile	2,239	Six years' teaching experience. "In manner gentle, in affection mild."
Other	34	No history; no write up.
Quotation / maxim	11,311	"As capricious as a woman"
Caption / fragment	1,363	Mary Pickford's only living rival.
Basic profile	650	His future vocation is journalism.

Model Architecture

Model	Architecture	Training / generation	Typical size	Primary use
Transformer	Encoder–Decoder	Encode source sequence; decode target sequence	Base $\approx$ 65M	Translation; sequence-to-sequence
BERT	Encoder-only	Bidirectional masked-language modeling	Base 110M; Large 340M	Understanding; classification; representations
GPT	Decoder-only	Autoregressive next-token prediction	GPT-1 117M; GPT-2 1.5B	Generation; completion; dialogue

Sources: Vaswani et al. (2017); Devlin et al. (2019); Radford et al. (2018); OpenAI GPT-2.

BERT Interpretation



LASSO and BERT

LASSO

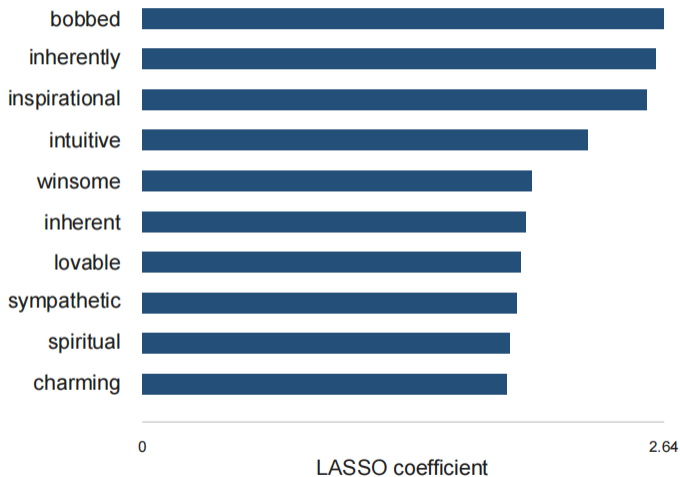
- Female = 1; male = 0
- Word presence (0/1); frequency ≥ 20
- Class-weighted logistic model
- Five-fold cross-validation
- Sign indicates gender association

BERT

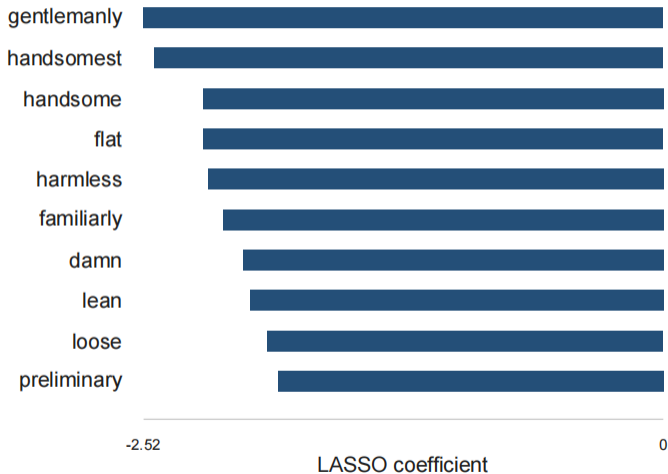
- bert-base-uncased; 110M parameters
- 192 tokens; two epochs
- Output: $P(\text{female})$
- Contribution: original minus masked
- Word mean; $n \geq 10$; BH $q < 0.05$

Masked: gender terms, names, capitalized words, institutions, fields, professions, military terms, sports, clubs, offices, and class years.

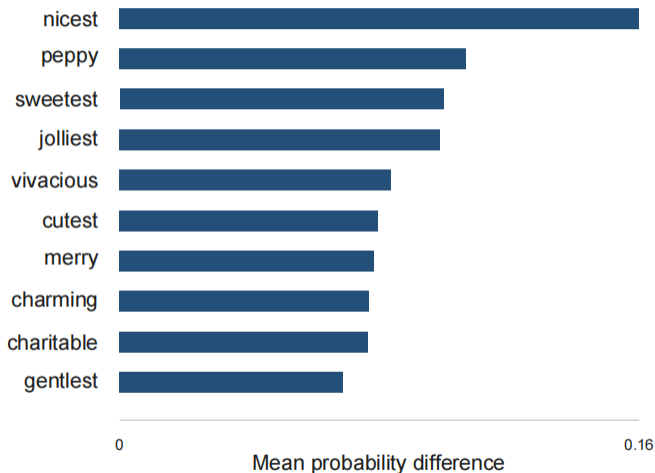
LASSO: Female-associated Words



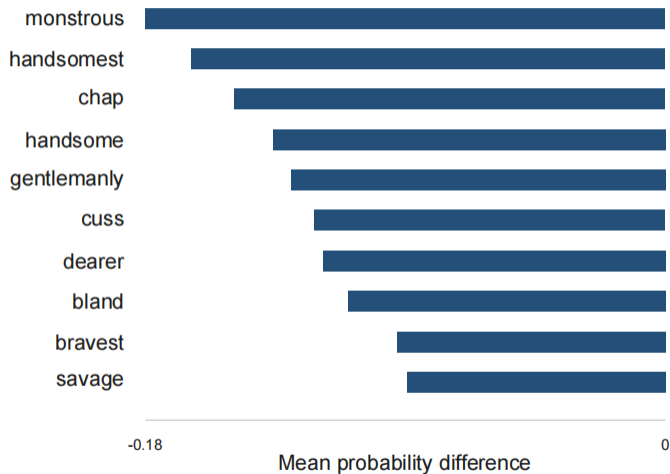
LASSO: Male-associated Words



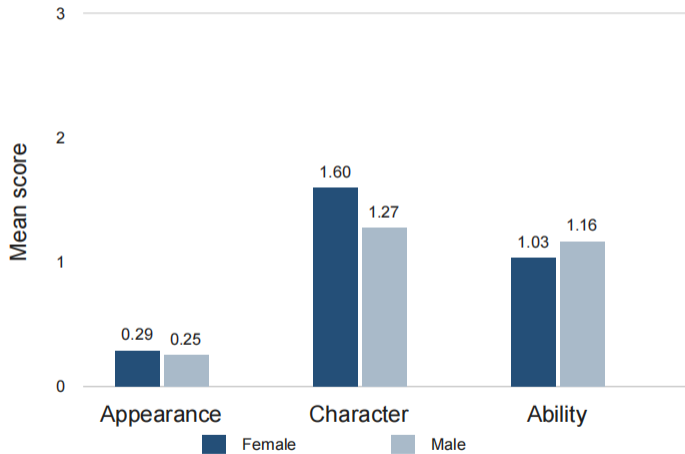
BERT: Female-associated Words



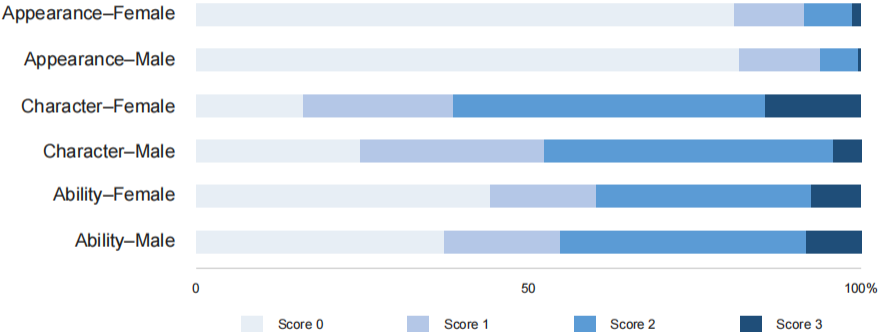
BERT: Male-associated Words



Qwen Scores



Qwen Score Distributions



Full Sample: Female Regressions

	Appearance	Character	Ability	LASSO bias	BERT bias
Male share	0.103*** (0.036)	0.154* (0.076)	0.174 (0.118)	0.019 (0.037)	0.050 (0.062)
share - 0.5	0.044 (0.056)	0.188 (0.160)	0.482** (0.205)	0.150 (0.092)	0.186** (0.087)
(share - 0.5) ²	0.097 (0.127)	0.397 (0.371)	0.817* (0.471)	0.346 (0.214)	0.387* (0.204)
Observations	8979	8979	8979	8979	8979
Year fixed effects	Yes	Yes	Yes	Yes	Yes
School fixed effects	Yes	Yes	Yes	Yes	Yes

Clustered standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Full Sample: Male Regressions

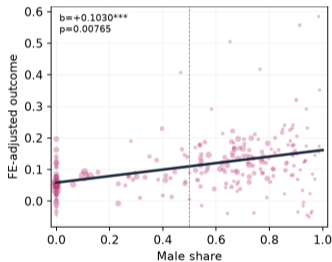
	Appearance	Character	Ability	LASSO bias	BERT bias
Male share	0.040 (0.055)	0.099 (0.092)	0.008 (0.116)	0.051 (0.040)	0.061 (0.046)
share - 0.5	0.044 (0.061)	0.090 (0.095)	0.042 (0.120)	0.065 (0.044)	0.091* (0.046)
(share - 0.5) ²	0.062 (0.104)	-0.066 (0.208)	-0.146 (0.234)	0.014 (0.092)	0.063 (0.077)
Observations	28459	28459	28459	28459	28459
Year fixed effects	Yes	Yes	Yes	Yes	Yes
School fixed effects	Yes	Yes	Yes	Yes	Yes

Clustered standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

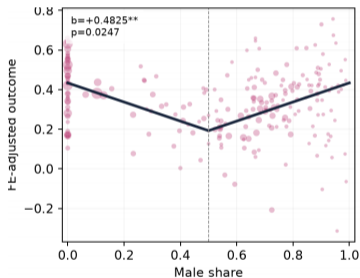
Gender Composition and Descriptions of Women

Full sample; female comments

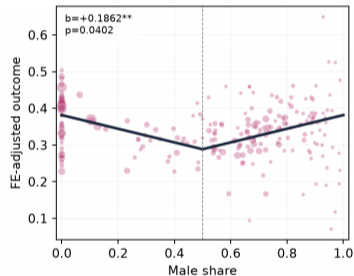
Appearance: linear



Ability: absolute deviation



BERT: absolute deviation



Shown: $p < 0.05$. Complete results: regression tables and appendix.

Text-Focused Sample: Female Regressions

	Appearance	Character	Ability	LASSO bias	BERT bias
Male share	0.105*** (0.036)	0.156** (0.076)	0.173 (0.118)	0.021 (0.037)	0.027 (0.044)
share - 0.5	0.050 (0.057)	0.188 (0.163)	0.491** (0.204)	0.141 (0.092)	0.164** (0.073)
(share - 0.5) ²	0.108 (0.131)	0.400 (0.379)	0.835* (0.471)	0.332 (0.212)	0.339** (0.166)
Observations	8887	8887	8887	8887	8887
Year fixed effects	Yes	Yes	Yes	Yes	Yes
School fixed effects	Yes	Yes	Yes	Yes	Yes

Clustered standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Text-Focused Sample: Male Regressions

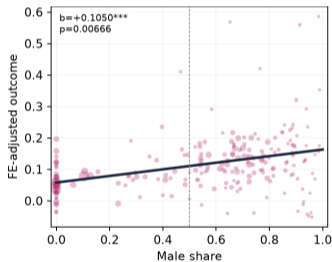
	Appearance	Character	Ability	LASSO bias	BERT bias
Male share	0.038 (0.057)	0.101 (0.092)	0.010 (0.117)	0.051 (0.039)	0.019 (0.031)
share - 0.5	0.044 (0.063)	0.098 (0.095)	0.045 (0.122)	0.067 (0.043)	0.037 (0.033)
(share - 0.5) ²	0.065 (0.105)	-0.047 (0.208)	-0.139 (0.239)	0.019 (0.091)	0.001 (0.053)
Observations	27902	27902	27902	27902	27902
Year fixed effects	Yes	Yes	Yes	Yes	Yes
School fixed effects	Yes	Yes	Yes	Yes	Yes

Clustered standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

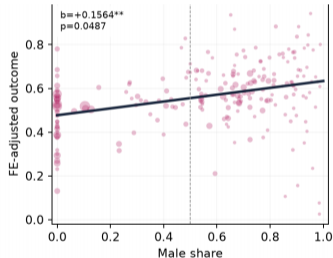
Gender Composition and Descriptions of Women

Text-focused sample; female comments; Qwen scores

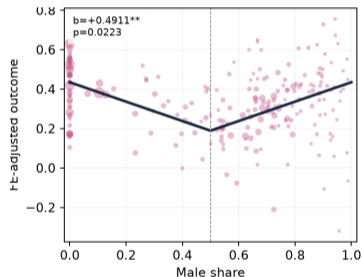
Appearance: linear



Character: linear



Ability: absolute deviation

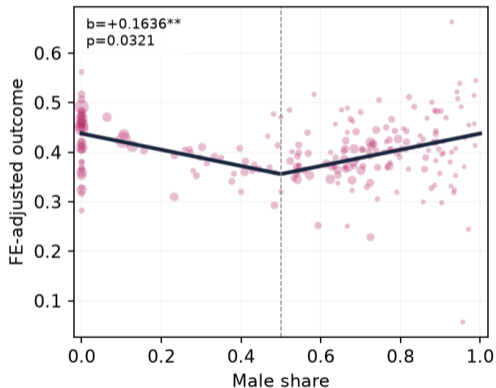


Shown: $p < 0.05$. Complete results: regression tables and appendix.

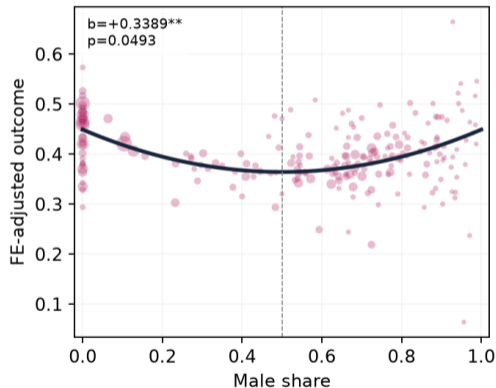
Gender Imbalance and Gendered Language

Text-focused sample; female comments; BERT score

BERT: absolute deviation



BERT: squared deviation



Shown: $p < 0.05$. Complete results: regression tables and appendix.

School-Year Means: Female Regressions

	Appearance	Character	Ability	LASSO bias	BERT bias
Male share	0.016 (0.068)	0.071 (0.125)	-0.018 (0.198)	-0.044 (0.058)	-0.082 (0.057)
share - 0.5	-0.043 (0.065)	-0.013 (0.159)	0.084 (0.240)	-0.036 (0.063)	-0.072 (0.069)
(share - 0.5) ²	-0.079 (0.167)	0.014 (0.321)	0.058 (0.533)	-0.071 (0.135)	-0.157 (0.158)
Observations	220	220	220	220	220
Year fixed effects	Yes	Yes	Yes	Yes	Yes
School fixed effects	Yes	Yes	Yes	Yes	Yes

Clustered standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

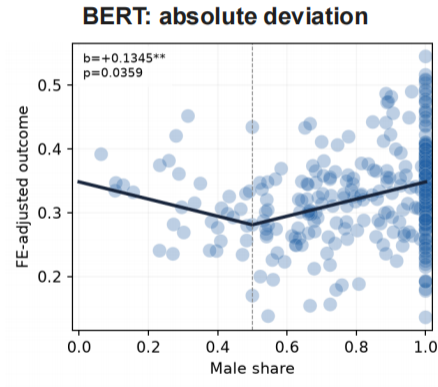
School-Year Means: Male Regressions

	Appearance	Character	Ability	LASSO bias	BERT bias
Male share	-0.016 (0.089)	0.057 (0.133)	-0.010 (0.144)	0.050 (0.056)	0.035 (0.071)
share - 0.5	0.051 (0.076)	0.059 (0.146)	0.059 (0.149)	0.091 (0.061)	0.135** (0.061)
(share - 0.5) ²	0.074 (0.132)	0.013 (0.276)	-0.007 (0.303)	0.124 (0.140)	0.191 (0.127)
Observations	355	355	355	355	355
Year fixed effects	Yes	Yes	Yes	Yes	Yes
School fixed effects	Yes	Yes	Yes	Yes	Yes

Clustered standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Gender Imbalance and Gendered Language

School-year means; male comments; BERT score



Shown: $p < 0.05$. Complete results: regression tables and appendix.

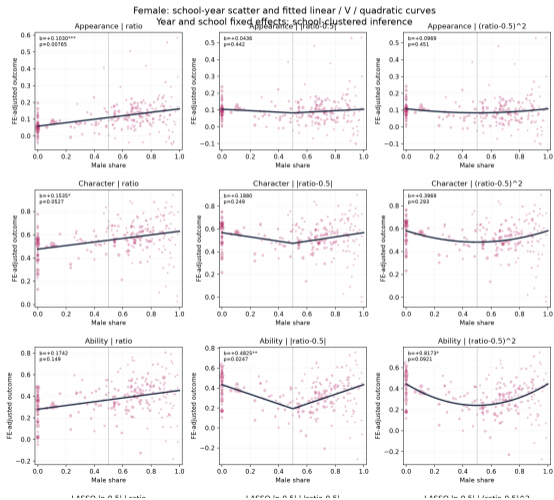
Appendix

Qwen Scores: Summary Table

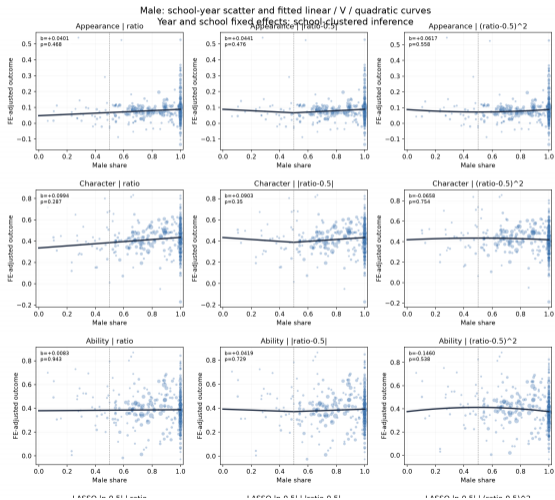
Sex	Appearance	Character	Ability
Female	0.29 (0.65)	1.60 (0.92)	1.03 (1.03)
Male	0.25 (0.57)	1.27 (0.88)	1.16 (1.02)

Cells report mean (standard deviation).

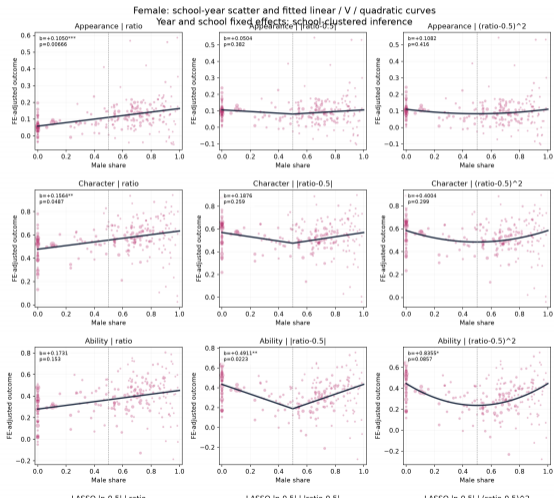
Full Sample: Female, Qwen



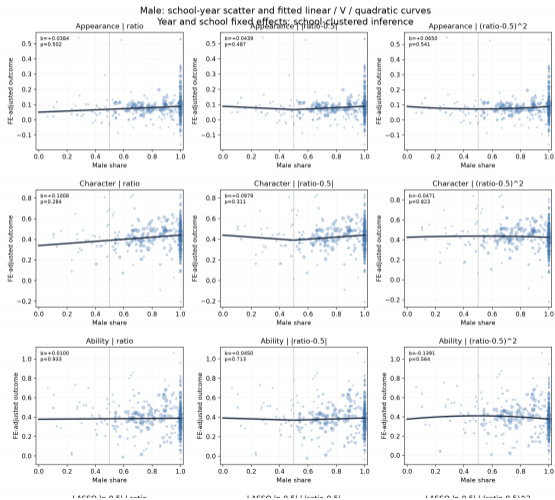
Full Sample: Male, Qwen



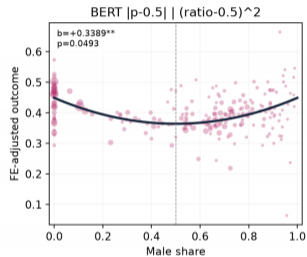
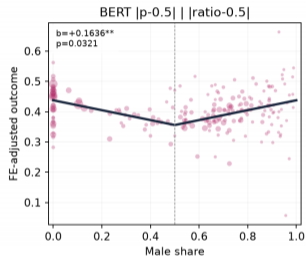
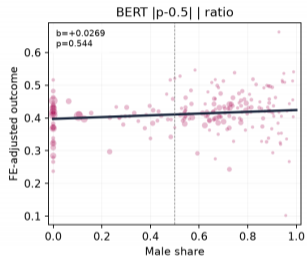
Text-Focused Sample: Female, Qwen



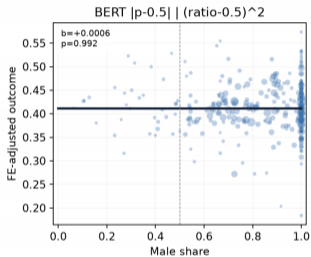
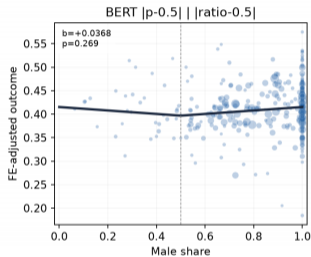
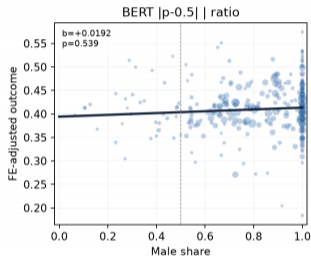
Text-Focused Sample: Male, Qwen



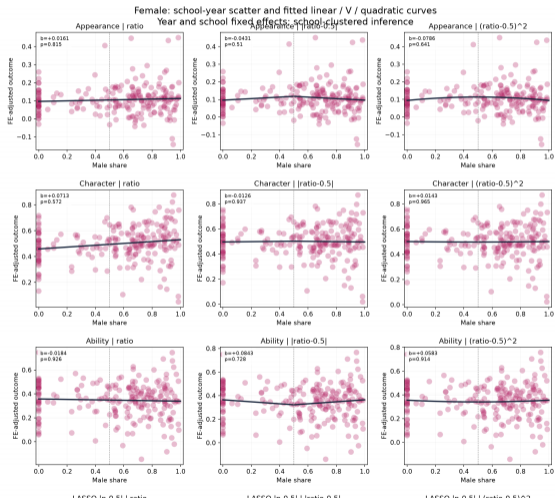
Text-Focused Sample: Female, BERT Bias



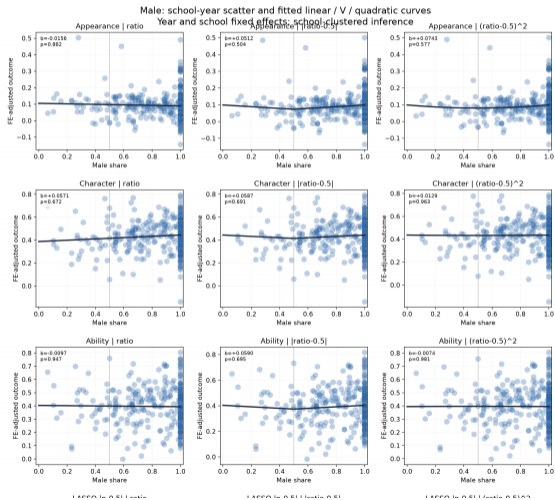
Text-Focused Sample: Male, BERT Bias



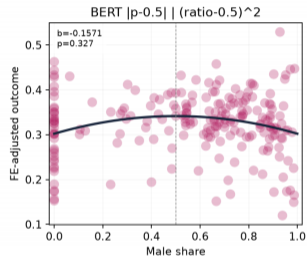
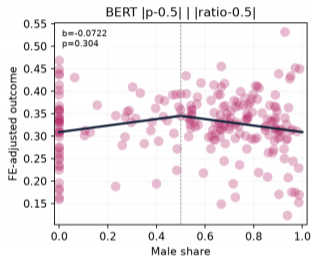
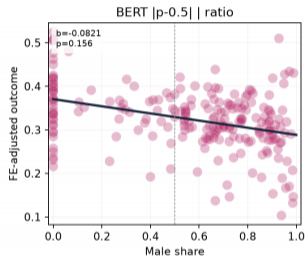
School-Year Means: Female, Qwen



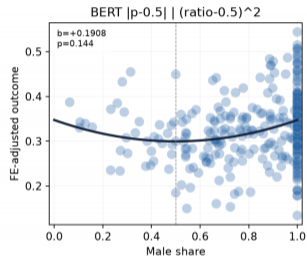
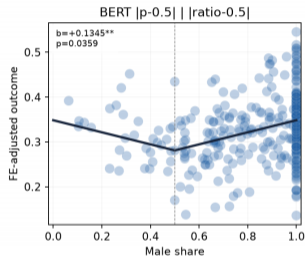
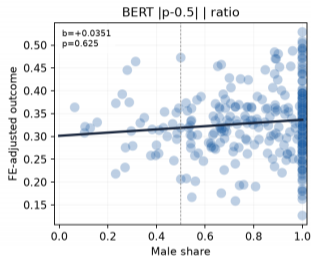
School-Year Means: Male, Qwen



School-Year Means: Female, BERT Bias



School-Year Means: Male, BERT Bias



LASSO: Word Statistics

Female-associated

Word	Coef.	Frequency
bobbed	2.645	22
inherently	2.603	77
inspirational	2.556	44
intuitive	2.258	39
winsome	1.972	70
inherent	1.943	142
lovable	1.918	215
sympathetic	1.896	170
spiritual	1.861	73
charming	1.846	246

Male-associated

Word	Coef.	Frequency
gentlemanly	-2.517	51
handsomest	-2.464	30
handsome	-2.229	290
flat	-2.228	39
harmless	-2.205	67
familiarly	-2.134	47
damn	-2.037	24
lean	-2.002	97
loose	-1.917	63
preliminary	-1.865	24

BERT: Word Statistics

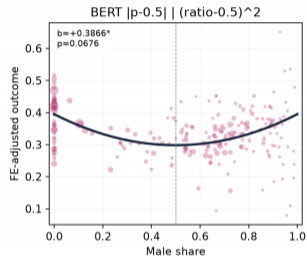
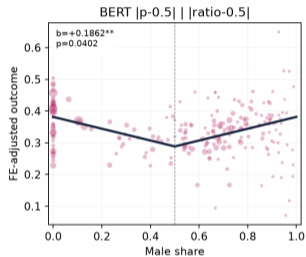
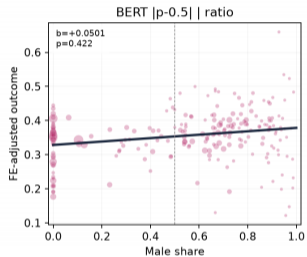
Female-associated

Word	Coef.	Frequency	<i>p</i> -value
nicest	0.159	13	0.001
peppy	0.106	13	0.001
sweetest	0.099	21	0.008
jolliest	0.098	21	0.004
vivacious	0.083	43	< 0.001
cutest	0.079	16	< 0.001
merry	0.078	191	< 0.001
charming	0.076	246	< 0.001
charitable	0.076	22	0.008
gentlest	0.068	20	< 0.001

Male-associated

Word	Coef.	Frequency	<i>p</i> -value
monstrous	-0.178	10	< 0.001
handsomest	-0.162	30	< 0.001
chap	-0.147	13	0.003
handsome	-0.134	290	< 0.001
gentlemanly	-0.128	51	< 0.001
cuss	-0.120	19	0.009
dearer	-0.117	10	< 0.001
bland	-0.109	11	0.007
bravest	-0.092	18	< 0.001
savage	-0.088	11	0.005

Full Sample: Female, BERT Bias



Full Sample: Male, BERT Bias

